InfyOne AI Assistant — Render.com Deployment Guide

Production Cloud Deployment, Service Configuration & Key Vault / Environment Variables

1. Deployment Architecture Overview

The application is containerized and cloud-ready for **Render.com Web Services**. It features a **FastAPI** backend running on Python, orchestrating a **LangGraph** stateful agent checkpointer, backed by **OpenAI (gpt-4o-mini & text-embedding-3-large)**, **Chroma Cloud** for vector knowledge retrieval, and optional **Redis Cloud** for distributed session persistence. All configuration is dynamically injected through Render's Environment Key Vault at container startup.

2. Render.com Web Service Setup Instructions

Follow these exact steps in your **Render Dashboard (dashboard.render.com)**:

1. **Create Web Service:** Click *New +* > *Web Service*, then connect your GitHub/GitLab repository.
2. **Runtime Environment:** Select **Python 3**.
3. **Build Command:** `pip install -r requirements.txt`
4. **Start Command:** `uvicorn app.main:app --host 0.0.0.0 --port $PORT`
5. **Health Check Path:** `/api/health` (ensures zero-downtime rolling deploys).
6. **Instance Plan:** Free or Starter (Standard 512MB RAM minimum).

3. Render Key Vault: Complete Environment Variables List

Enter the following Key-Value pairs in your Render Dashboard under **Environment > Environment Variables**. Render securely encrypts these keys and injects them directly into the runtime container.

Category	Key Vault Name	Required ?	Production Value / Setting	Purpose & Description
Application	`APP_NAME`	Optional	`InfyOne`	Public application name displayed in logs & health check.
Application	`APP_ENV`	Required	`production`	Disables debug overhead, swagger docs, and activates production handlers.
Application	`APP_DEBUG`	Required	`false`	Enforces strict production security and minimizes verbose tracebacks.
Application	`APP_HOST`	Required	`0.0.0.0`	Binds FastAPI to all network interfaces inside the Render container.
Application	`APP_ALLOWED_HOSTS`	Required	`["*"]`	Allows requests routed via *.onrender.com and custom domain names.
Application	`APP_CORS_ORIGINS`	Required	`["*"]`	Enables browser web chat client cross-origin API communication.
Application	`APP_SECRET_KEY`	Required	`0fc53de77df447e9dd4622...`	Cryptographic secret token for session signatures and security.
OpenAI LLM	`OPENAI_API_KEY`	CRITICAL	`sk-proj-K6--23M... (Your Key)`	OpenAI API authentication key for generation and embeddings.
OpenAI LLM	`OPENAI_MODEL`	Required	`gpt-4o-mini`	Fast, low-latency, highly accurate model configured for chat pipeline.
Embeddings	`EMBEDDING_PROVIDER`	Required	`openai`	Embedding generator engine (OpenAI text-embedding API).
Embeddings	`EMBEDDING_MODEL`	Required	`text-embedding-3-large`	3072-dimensional embedding model matching the Chroma collection.
Embeddings	`EMBEDDING_DIMENSION`	Required	`3072`	Vector dimension specification for cosine distance similarity.

Category	Key Vault Name	Required ?	Production Value / Setting	Purpose & Description
Chroma Cloud	`CHROMA_CLOUD_API_KEY`	CRITICAL	`ck-CLbEQVRCf58NnFX92...`	API key connecting backend directly to hosted Chroma Cloud vector DB.
Chroma Cloud	`CHROMA_TENANT`	Required	`da86004b-97e5-490c-ac...`	Chroma Cloud account tenant identifier (UUID format).
Chroma Cloud	`CHROMA_DATABASE`	Required	`StarkBots`	Chroma Cloud database holding Infosys knowledge chunks.
Chroma Cloud	`CHROMA_COLLECTION_NAME`	Required	`InfyOne`	Target collection containing the 100 Infosys knowledge chunks.
Chroma Cloud	`CHROMA_DISTANCE_METRIC`	Required	`cosine`	Distance metric used for HNSW vector space similarity search.
Redis Cache	`REDIS_HOST`	Optional	`redis-14271.c330...redislabs.com`	Hosted Redis endpoint for cross-restart distributed caching.
Redis Cache	`REDIS_PORT`	Optional	`14271`	Redis Cloud service network port number.
Redis Cache	`REDIS_PASSWORD`	Optional	`IZ3rUdsemHJsoiM3V9...`	Redis authentication password credential.
Logging	`LOG_LEVEL`	Optional	`INFO`	Terminal log verbosity level (INFO recommended in production).
Logging	`LOG_FORMAT`	Optional	`json`	Formats output as single-line JSON records for Render log viewer.

4. Post-Deployment Verification Checklist

Once Render deploys your service, test these 3 endpoints to verify operation:

1. **Health Check:** `GET https://<your-app>.onrender.com/api/health` → returns `{"status": "healthy"}`.
2. **Web Interface:** Visit `https://<your-app>.onrender.com/` → loads interactive InfyOne chatbot UI.
3. **Chat Execution:** Send `POST /api/chat` with payload `{"message": "Who is the next Infosys CEO?"}` → returns Ashiss Kumar Dash + sources.

Render Blueprint File: A pre-configured `render.yaml` has also been created in your repository root for 1-click automated Blueprint deployment.